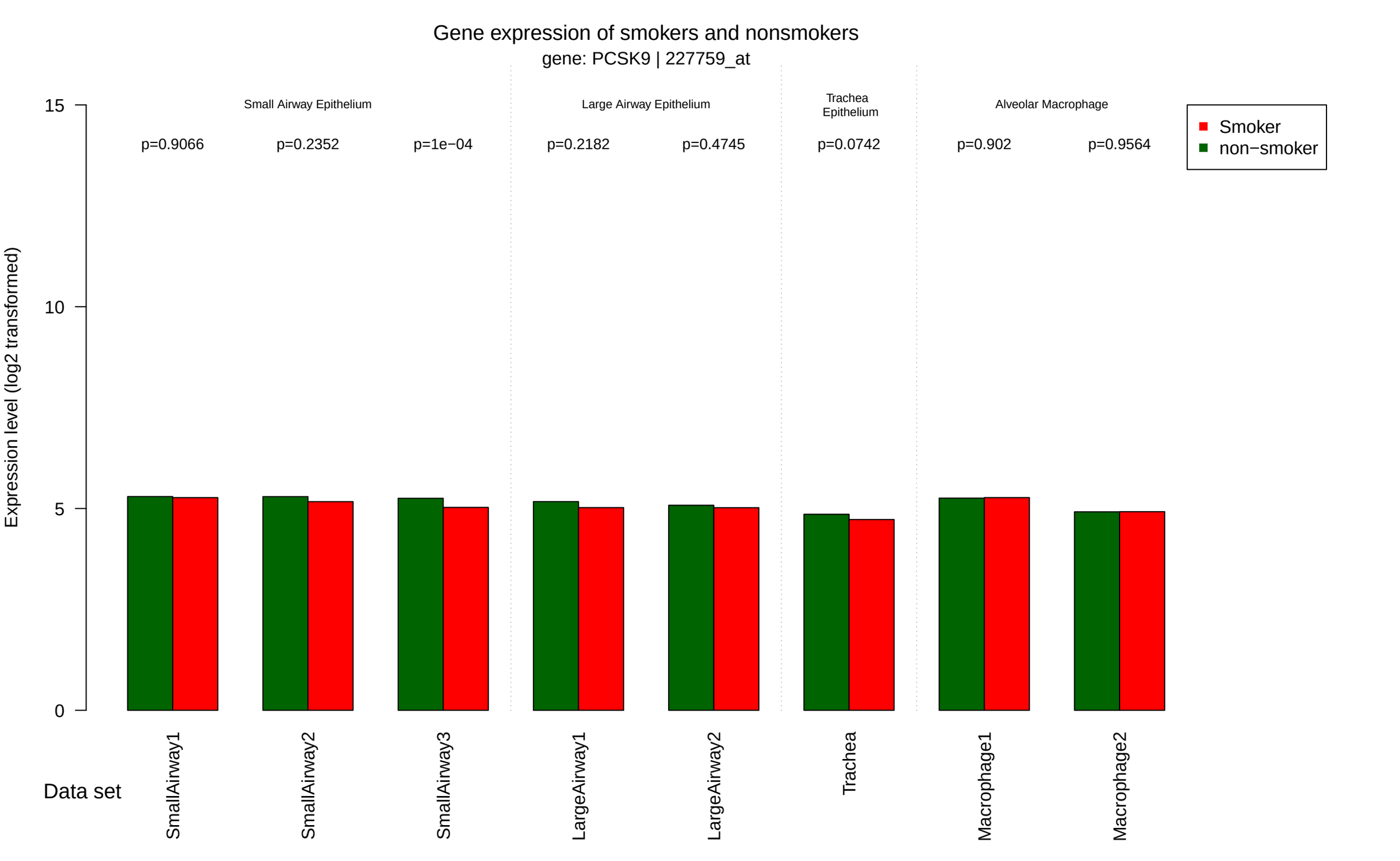
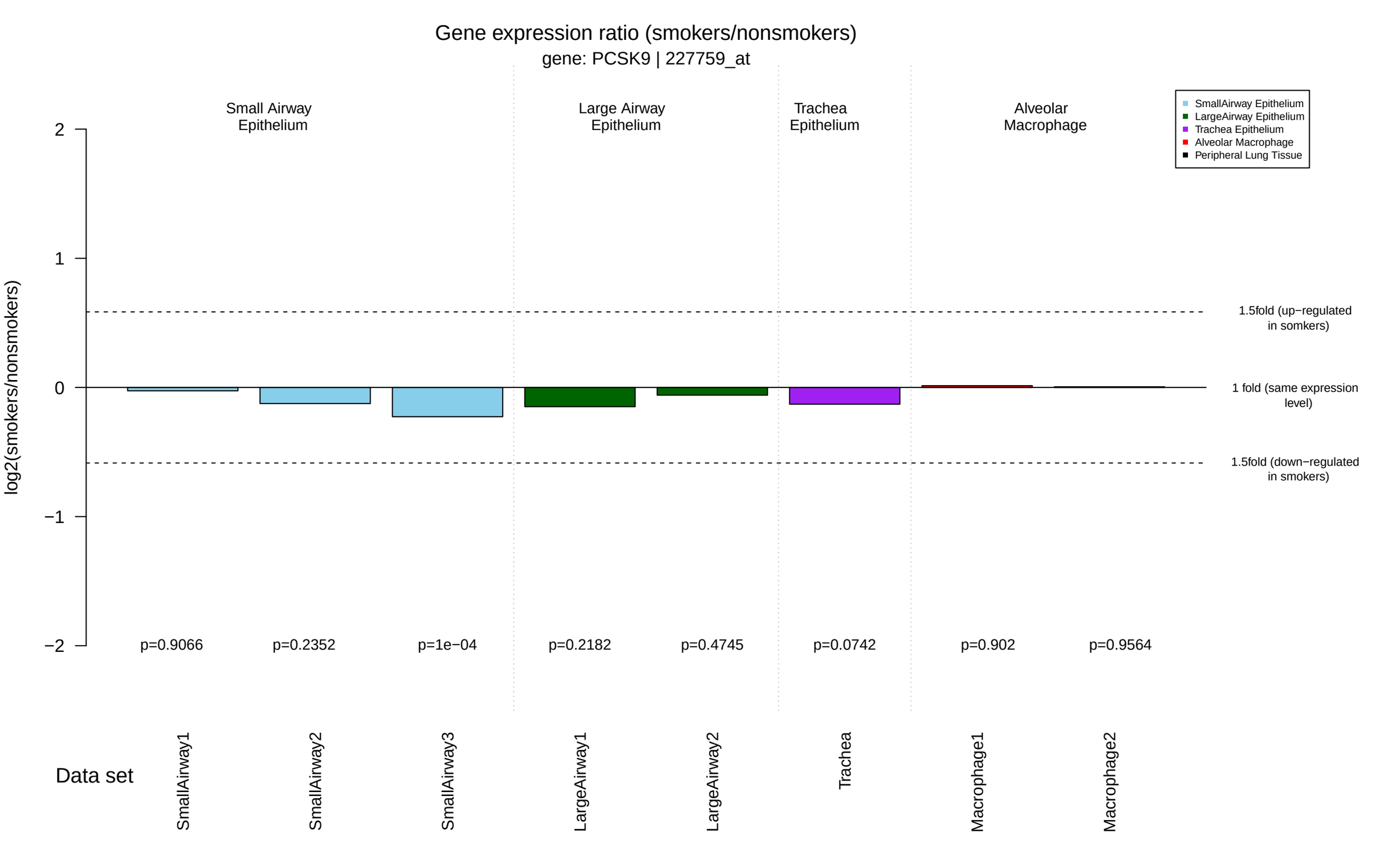
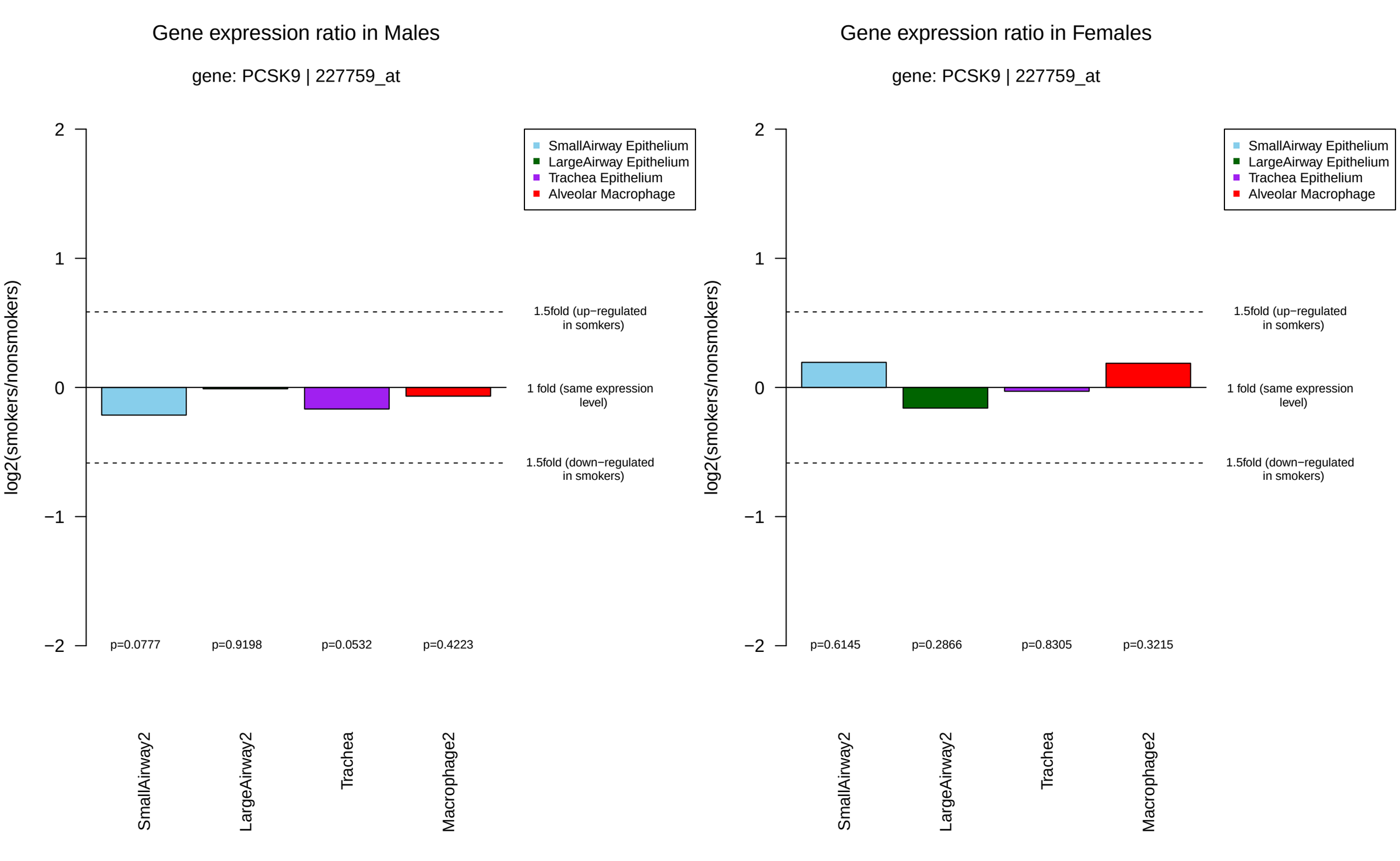
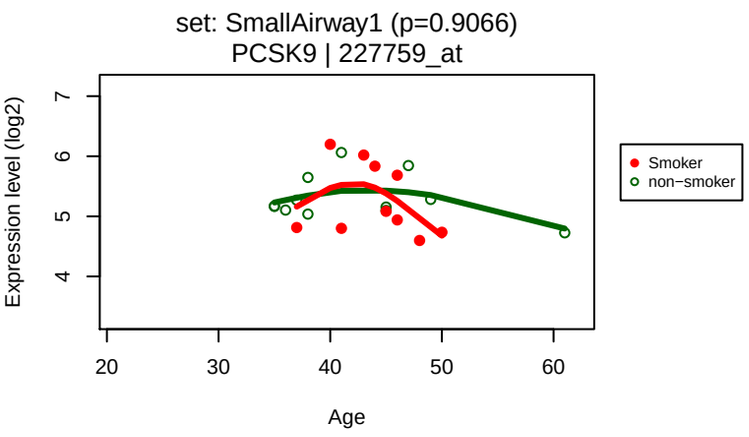
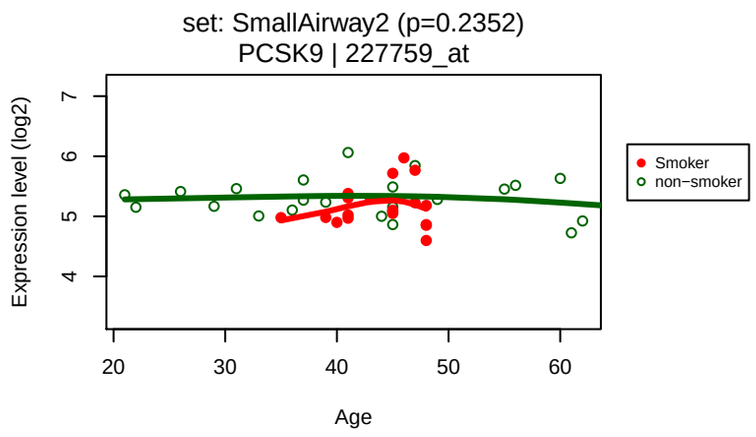
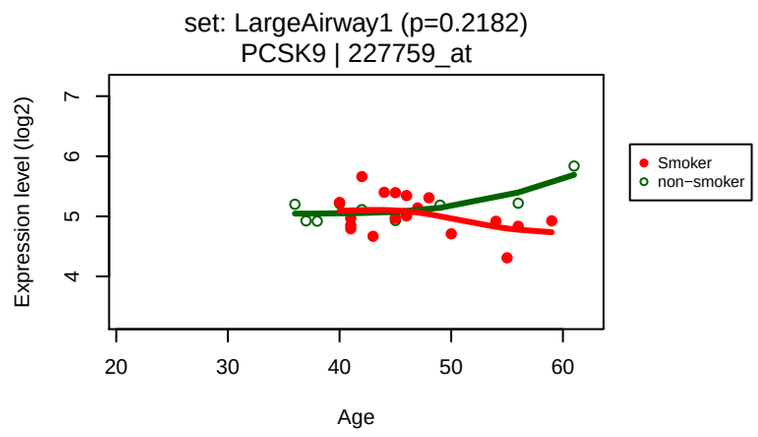

Gene expression of smokers and nonsmokers
gene: PCSK9 | 227759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9066
p=0.2352
p=1e−04
p=0.2182
p=0.4745
p=0.0742
p=0.902
p=0.9564
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

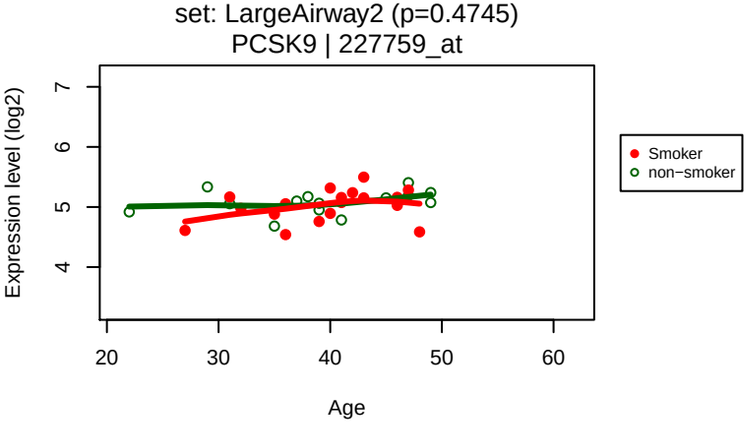

Gene expression ratio (smokers/nonsmokers)
gene: PCSK9 | 227759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9066
p=0.2352
p=1e−04
p=0.2182
p=0.4745
p=0.0742
p=0.902
p=0.9564
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

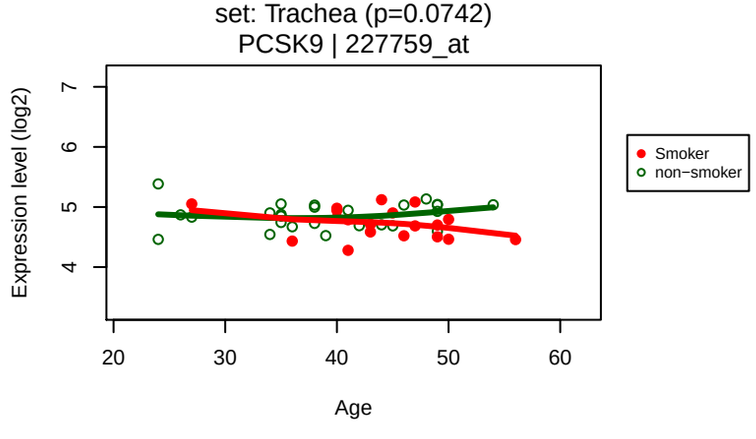

Gene expression ratio in Males
gene: PCSK9 | 227759_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0777
p=0.9198
p=0.0532
p=0.4223
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PCSK9 | 227759_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6145
p=0.2866
p=0.8305
p=0.3215
SmallAirway2
LargeAirway2
Trachea
Macrophage2

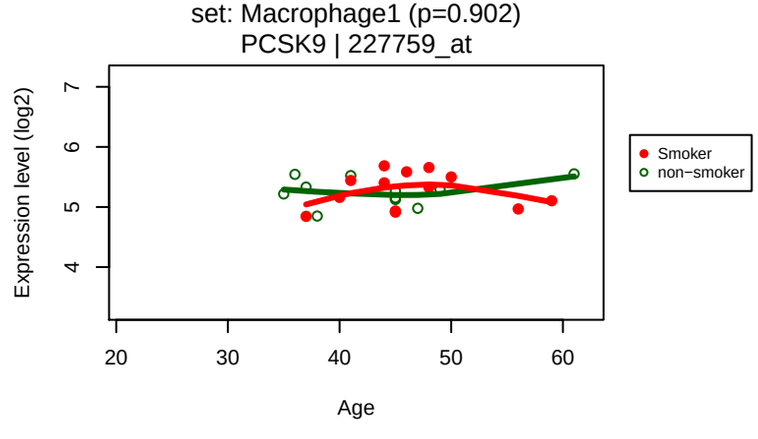

set: SmallAirway1 (p=0.9066)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker

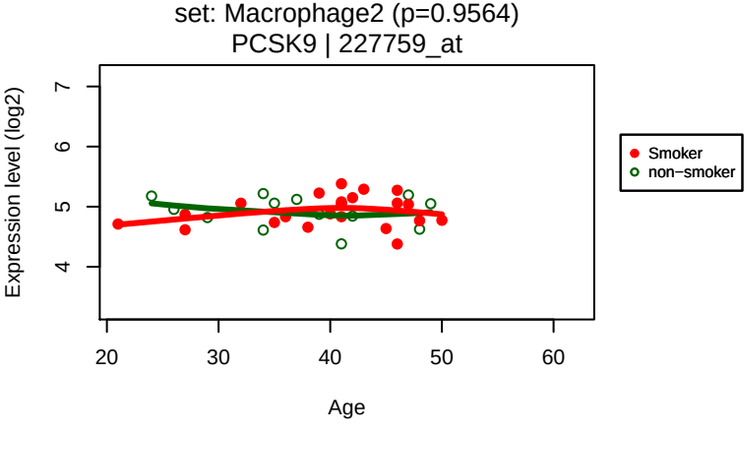

set: SmallAirway2 (p=0.2352)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker

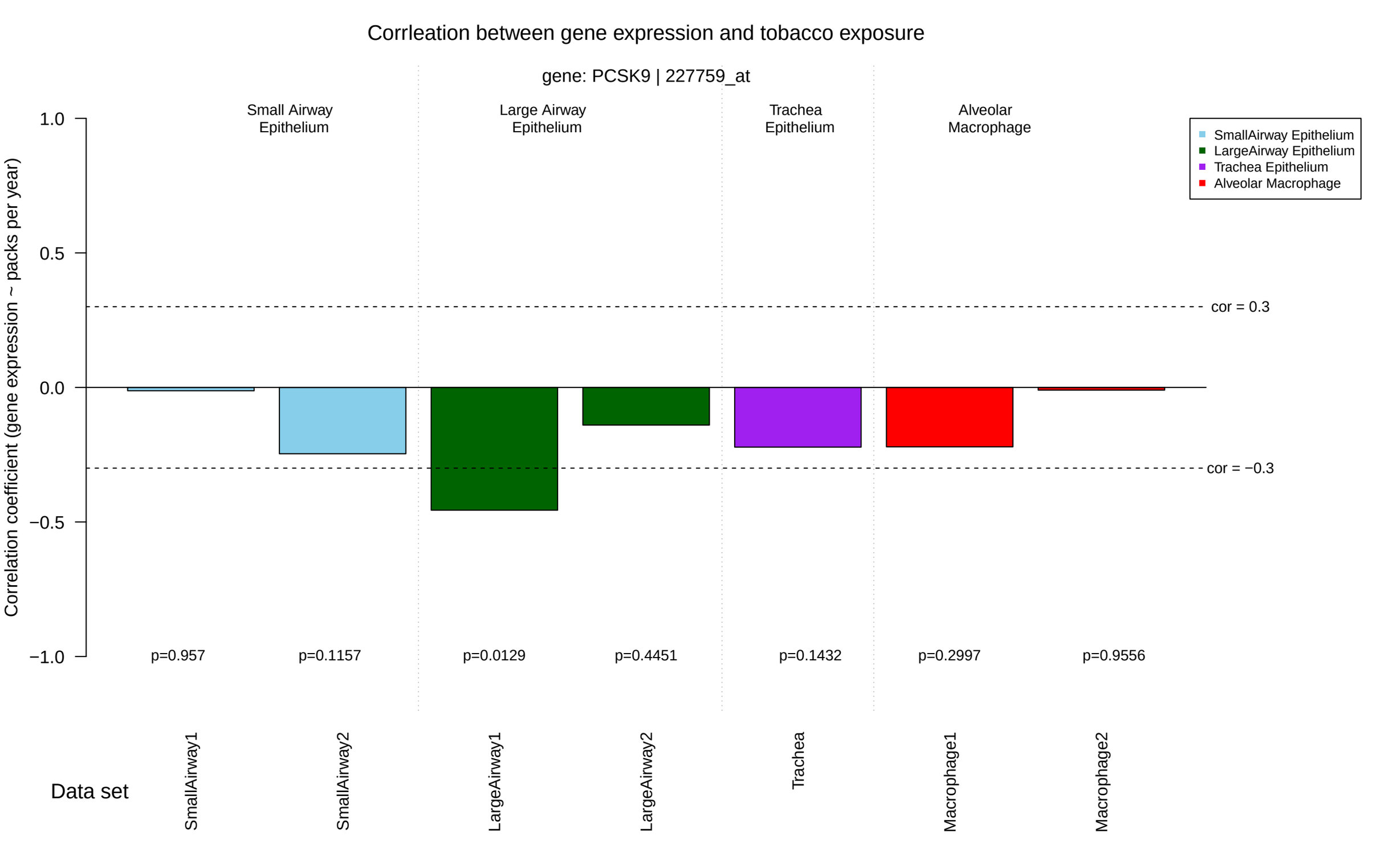

set: LargeAirway1 (p=0.2182)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4745)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0742)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.902)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9564)
PCSK9 | 227759_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PCSK9 | 227759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.957
p=0.1157
p=0.0129
p=0.4451
p=0.1432
p=0.2997
p=0.9556
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2